



Safety Data Sheet

Preparation Date 10-May-2015

Revision date 20-Nov-2024

Revision Number: 7

1. Identification of the Substance/Preparation and of the Company/Undertaking

Identification of the product

Product Description Metricor 4F

Other means of identification

Internal SDS code 12U-145Z

Registration number(s) 70506-68

Recommended use of the chemical and restrictions on use

Recommended use Herbicide.

Uses advised against Activities contrary to label recommendation

Details of the Supplier of the Safety Data Sheet

Supplier Address

UPL NA Inc.

PO Box 12219

Research Triangle Park, NC 27709-12219

Emergency telephone number

Company Phone Number 1-800-438-6071

Emergency telephone number Chemtrec: (800) 424-9300 (24hrs) or (703) 527-3887
Medical: Rocky Mountain Poison and Drug Safety (866) 673-6671 (24hrs)

2. Hazards Identification

Classification

OSHA Regulatory Status

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Acute toxicity - Oral	Category 4
Acute toxicity - Inhalation (Dusts/Mists)	Category 4
Specific target organ toxicity (repeated exposure)	Category 2

Label elements

EMERGENCY OVERVIEW

WARNING

Hazard Statements

HARMFUL IF SWALLOWED

Harmful if inhaled

Causes damage to organs through prolonged or repeated exposure
(Blood)



Appearance suspension**Physical state** Viscous Liquid**Odor** faint**Precautionary Statements - Prevention**

Do not breathe dust/fume/gas/mist/vapors/spray
Use only outdoors or in a well-ventilated area
Wash face, hands and any exposed skin thoroughly after handling
Wear protective gloves/protective clothing/eye protection/face protection
Do not eat, drink or smoke when using this product

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing
Immediately call a POISON CENTER or doctor/physician
IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell
Rinse mouth

Precautionary Statements - Storage

Store in a well-ventilated place. Keep container tightly closed
Store locked up

Precautionary Statements - Disposal

Dispose of contents/container to an approved waste disposal plant

Hazards Not Otherwise Classified (HNOC)**OTHER INFORMATION**

- Toxic to aquatic life

3. Composition/information on Ingredients

Chemical name	CAS No	Weight-%
Metribuzin technical	21087-64-9	41
Propylene Glycol	57-55-6	<15

*The exact percentage (concentration) of composition has been withheld as a trade secret.

4. First aid measures**FIRST AID MEASURES**

Eye contact	Hold eye open and rinse slowly and gently with water for 15 - 20 minutes. Remove contact lenses, if present, after 5 minutes, then continue rinsing eye. Call a poison control center or doctor for treatment advice.
Skin contact	Wash off immediately with plenty of water for at least 15 minutes. Call a poison control center or doctor for treatment advice.
Inhalation	Move person to fresh air. If person is not breathing, call 911 or an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. Call a physician or poison control center immediately.
Ingestion	Immediate medical attention is not required. Rinse mouth. Drink plenty of water. If symptoms persist, call a physician. Do not induce vomiting without medical advice. Never give anything by mouth to an unconscious person. Call a physician.
Protection of First-aiders	Use personal protective equipment.

Most Important Symptoms and Effects, Both Acute and Delayed

Most Important Symptoms and May cause damage to organs through prolonged or repeated exposure. (Blood effects).

Effects**Indication of Any Immediate Medical Attention and Special Treatment Needed**

Notes to physician No information available.

5. Fire-fighting measures**Suitable extinguishing media**

Carbon dioxide (CO₂). Aquatic. Foam.

Unsuitable extinguishing media no data available.

Specific hazards arising from the chemical

Thermal decomposition can lead to release of irritating gases and vapours.

Hazardous combustion products Oxides of nitrogen. Oxides of carbon. Hydrogen cyanide.

Explosion data**Protective equipment and precautions for firefighters**

Use personal protective equipment. As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

6. Accidental release measures**Personal precautions, protective equipment and emergency procedures****Personal Precautions**

Avoid contact with skin and eyes. Wear protective gloves/protective clothing and eye/face protection. Wash thoroughly after handling. Stop leak if you can do it without risk. If not wearing an appropriate personal protective clothing, do not touch the damaged container and leakage. Wash clothing before reuse. Do not eat, drink or smoke when using this product.

Environmental Precautions**Environmental precautions**

Prevent further leakage or spillage if safe to do so. Consult a regulatory specialist to determine appropriate state or local reporting requirements, for assistance in waste characterization and/or hazardous waste disposal and other requirements listed in pertinent environmental permits.

Methods and material for containment and cleaning up**Methods for Clean-Up**

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Sweep up and shovel into suitable containers for disposal.

7. Handling and Storage**Precautions for safe handling****Handling**

Do not eat, drink or smoke when using this product. Avoid contact with skin and eyes. Wear personal protective equipment. Remove and wash contaminated clothing before re-use. Do not breathe vapours or spray mist. Provide adequate ventilation. Wash thoroughly after handling. Keep out of reach of children.

Conditions for safe storage, including any incompatibilities**Storage**

Keep in a dry place. Keep containers tightly closed in a cool, well-ventilated place. Keep in properly labelled containers. Keep out of the reach of children.

incompatible materials

Strong oxidizing agents.

8. Exposure Controls/Personal Protection

Exposure guidelines

Chemical name	ACGIH TLV	OSHA PEL
Metribuzin technical	TWA: 5 mg/m ³	(vacated) TWA: 5 mg/m ³

Engineering controls Investigate engineering techniques to reduce exposures. Local mechanical exhaust ventilation is preferred. Consult ACGIH ventilation manual or NFPA Standard 91 for design of exhaust systems.

Personal protective equipment**Eye/Face Protection**

Use eye protection to avoid eye contact. Where there is potential for eye contact have eye flushing equipment available. Safety glasses with side-shields.

Skin protection

Wear protective gloves/clothing. Socks and footwear.

Respiratory protection

Where airborne exposure is likely, use NIOSH approved respiratory protection equipment appropriate to the material and/or its components. Full facepiece equipment is recommended and, if used, replaces need for face shield and/or chemical goggles. If exposures cannot be kept at a minimum with engineering controls, consult respirator manufacturer to determine appropriate type equipment for given application. Observe respirator use limitations specified by NIOSH or the manufacturer. For emergency and other conditions where there may be a potential for significant exposure, use an approved full face positive-pressure, self-contained breathing apparatus. Respiratory protection programs must comply with 29 CFR 1910.134.

General hygiene considerations

Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing before re-use. Wear suitable gloves and eye/face protection. Wash hands before breaks and immediately after handling the product.

9. Physical and Chemical Properties

9.1 Information on basic physical and chemical properties

Appearance	suspension
Physical state	Viscous Liquid
Odor	faint
color	Off-white to brown-gray

Property	VALUES	Remarks/ Method
pH	7.37	None known
Melting point/freezing point	8.11 °C	None known
Boiling Point/Range		None known
Flash Point	> 100 °C	None known
Evaporation Rate		None known
Flammability (solid, gas)		None known
Specific gravity	1.16 g/cm ³ 20 c	None known
Bulk density		None known
Water solubility		Miscible
Solubility in Other Solvents	No information available	None known
Partition coefficient: n-octanol/water		None known
Autoignition temperature		None known
Decomposition temperature		None known
Viscosity		None known

9.2 OTHER INFORMATION- NONE

10. Stability and Reactivity

Reactivity

no data available

Chemical stability

Stable under normal conditions.

Possibility of hazardous reactions

None under normal processing.

Hazardous polymerization

Hazardous polymerisation does not occur.

Conditions to avoid

Sustained temperatures above 100 F.

Incompatible materials

Strong oxidizing agents.

Hazardous decomposition products

Carbon oxides. Amines. Methyl mercaptans.

11. Toxicological Information

Information on Likely Routes of Exposure**Product information**

LQ- 4F

Acute oral LD50 (rat) 1871 mg/kg

Acute dermal LD50 (rat) >2000 mg/kg No deaths

Acute inhalation LC50 (rat) >1.881 mg/L/4hr (aerosol) No deaths

Inhalation

Harmful if inhaled.

Eye contact

May cause irritation.

Skin contact

Based on available data, the classification criteria are not met.

Ingestion

HARMFUL IF SWALLOWED.

Information on Toxicological Effects**Symptoms**

No information available.

Delayed and immediate effects as well as chronic effects from short and long-term exposure**Sensitization**

No information available.

Mutagenic effects

no data available.

Carcinogenicity

There are no known carcinogenic chemicals in this product.

Reproductive effects

Not Available.

STOT - Single Exposure

no data available.

STOT - Repeated Exposure

Causes damage to organs through prolonged or repeated exposure.

Target organ effects

liver, blood, Thyroid, Central Nervous System (CNS).

Aspiration hazard

No information available.

Numerical Measures of Toxicity - Product information**LD50 Oral**

1871 mg/kg (rat)

LD50 Dermal

> 2000 mg/kg (rat)

LC50 Inhalation

> 1.881 mg/l (rat) No deaths

12. Ecological Information

ecotoxicity**LQ- 4F**

FISH LC50 (rainbow trout) 74.6 mg/l 96hr

AQUATIC INVERTEBRATES (Daphnia magna) 49.6 mg/l 48 hr

AQUATIC PLANTS (Green Algae) 72 hr 0.022 mg/l

Persistence/Degradability

no data available.

Bioaccumulation/ Accumulation

Does not bioaccumulate.

Other Adverse Effects

no data available

13. Disposal Considerations

Waste Treatment Methods**Waste Disposal Method**

Pesticide wastes can be hazardous. Improper disposal of excess pesticide or rinsate is a violation of Federal law. If the wastes cannot be disposed of by use or according to label instructions, contact your State Pesticide or Environmental Control Agency, or the Hazardous Waste representative at the nearest EPA Regional Office for guidance.

Contaminated packaging

Refer to product label.

14. Transport Information

DOT

When shipped domestically by highway in non-bulk containers this product can be shipped as not regulated.

When shipped in bulk, by vessel, or internationally refer to the IMDG shipping description:

TDG

When shipped in Canada domestic highway non-bulk this product can be shipped as Not regulated as per TDG 1.45.1

In bulk, by vessel or internationally - use IMDG description

IATA

UN/ID no
Proper shipping name
Hazard class
Packing group
Description

Single or inner packagings less than 5L (liquid) or 5kg net (solids) are excepted from the Dangerous Goods regulations (IATA Special provision A197)

UN3082
Environmentally hazardous substances, liquid, n.o.s (Metribuzin)
9
PG III
Marine pollutant

IMDG

UN/ID no
Proper shipping name
Hazard class
Packing group
Environmental hazards

Single or inner packaging less than 5L (liquid) or 5 kg net (solids) are excepted from the Dangerous Goods regulations (IMDG 2.10.2.7)

UN3082
Environmentally hazardous substances, liquid, n.o.s (metribuzin)
9
PG III
Marine pollutant

15. Regulatory Information

This chemical is a pesticide product registered by the Environmental Protection Agency and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets, and for workplace labels of non-pesticide chemicals. Following is the hazard information as required on the pesticide label:

signal word CAUTION

Ventilation Control PESTICIDE APPLICATORS & WORKERS THESE WORKERS MUST REFER TO PRODUCT LABELING AND DIRECTIONS FOR USE IN ACCORDANCE WITH EPA WORKER PROTECTION STANDARD 40 CFR PART 170.

Harmful if swallowed or absorbed through the skin. Keep out of Reach of Children.

International Inventories

USINV	Present
DSL/NDL	Not present
EINECS/	Not Present
ELINCS	
ENCS	Not Present
China	Present
KECL	Not Present
PICCS	Not Present
AICS	Not Present
TSCA	Not Present

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

ENCS - Japan Existing and New Chemical Substances

IECSC - China Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

AICS - Australian Inventory of Chemical Substances

Federal Regulations

SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372:

CERCLA

Not applicable

CERCLA

RCRA

Pesticide Information

Component	FIFRA - Restricted Use	FIFRA - Pesticide Product Other Ingredients	FIFRA - Listing of Pesticide Chemicals	California Pesticides - Restricted Materials
Metribuzin technical 21087-64-9 (41)			X	
Propylene Glycol 57-55-6 (<15)			X	

State Regulations

State Right-to-Know

Chemical name	Massachusetts	New Jersey	Pennsylvania	Illinois	Rhode Island
Metribuzin technical -	X	X	X		

21087-64-9					
Propylene Glycol - 57-55-6		X	X		

International regulations

U.S. EPA Label information

EPA Pesticide registration number 70506-68

16. Other Information

NFPA **HEALTH 1** **flammability 1** **Instability 0** **Physical hazard -**

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Revision Summary

Update section 2 Update section 4 Update section 11

Disclaimer

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End of SDS